EE332 Introduction to VLSI—Midterm

Spring 2008

Editable reconstruction of original pages 1-1 and 1-2. Study ink is excluded. The first four printed allocations total 90 points; Q5 is printed "10%" and is retained that way.

1. (40 points) Answer the following problems.

(a) What is antenna effect? (4 points)

(b) What is channel length modulation effect? (4 points)

(c) Draw a CMOS transistor-level schematic realizing $f = \overline{xy + (u+v)w}$. (6 points)

(d) List four properties of static CMOS logic. (4 points)

(e) List two latchup prevention approaches. (4 points)

(f) Draw a CMOS inverter and label the drain and source of the PMOS and NMOS transistors. (4 points)

(g) Define the low and high noise margins using the voltage transfer curve of a CMOS inverter. (4 points)

(h) List two possible approaches for alleviating the effect of crosstalk on two neighboring parallel interconnects. (4 points)

(i) List the three major components of the load capacitance of a CMOS logic gate. (6 points)

2. (20 points)

Figure 1(a) depicts a CMOS logic circuit. Assume that the delay of the inverter is 1 delay unit, the inverter size is $W_p/L_p = 2\,\mu\text{m}/0.18\,\mu\text{m}$ and $W_n/L_n = 1\,\mu\text{m}/0.18\,\mu\text{m}$, and its input capacitance is C_i. Also, assume that the size of each NMOS and PMOS transistor of all the other gates is the same as that of the inverter.

(a) Determine the delay of each gate in terms of delay unit (neglect the diffusion and wire capacitance). (10 points)

(b) Assume that a dual-V_t process can be used to implement the CMOS circuit. Draw the transistor schematic diagram in terms of the symbols shown in Fig. 1(b) such that leakage power is minimal without sacrificing the speed of the CMOS circuit. (10 points)

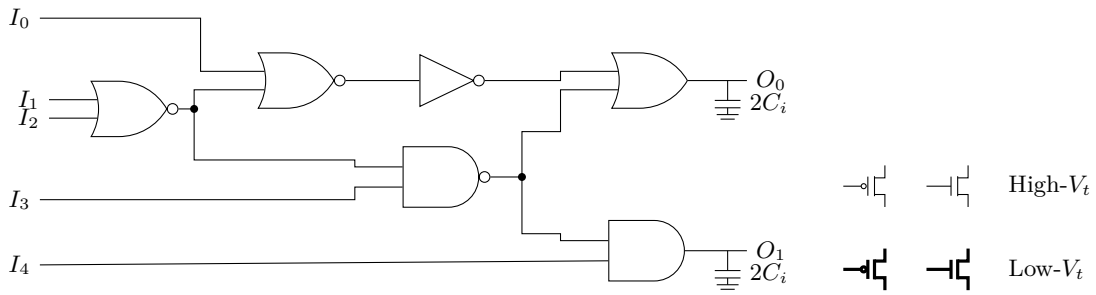


Figure 1: (a) A CMOS circuit (b) High-/low-V_t transistor symbols.

3. (20 points)

Consider a buffer chain shown in Fig. 2, used to drive a large capacitance C_L. The first-stage buffer is minimum size with input capacitance C_i. Each buffer has transistor width larger than the previous one by a factor β. Neglect parasitic capacitance. **(a) (10 points)** What are the size ratio β and number of stages N such that the delay of the buffer chain is minimal?

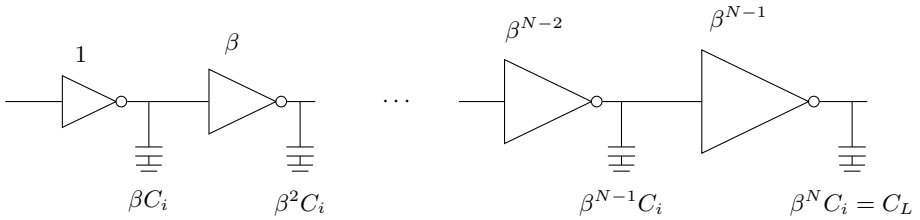


Figure 2: A taper buffer for driving a large capacitance.

3(b). (10 points) Calculate the power efficiency η, where $\eta = (P_T - P_L)/P_T$, when the optimized ratio is used to design the buffer chain. Here P_T and P_L denote total switching power of the buffer chain and load capacitance C_L, respectively.

4. (10 points)

The figure in Fig. 3 depicts an inverter chain driving a capacitive load C_L. Every inverter has a minimum-size NMOS and the PMOS/NMOS ratio is α. Let $\mu_n = 2\mu_p$ and let the resistance and input capacitance of the minimum-size NMOS be R and C. Derive α such that $T = 6RC$ and the power consumption constituted by the first two inverters is minimal.

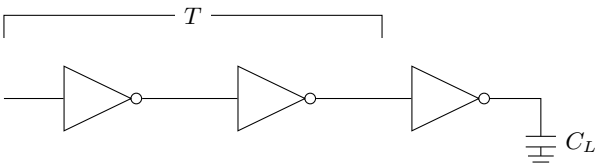


Figure 3: An inverter pair. (The printed figure contains three inverters; T brackets the first two.)

5. (10%)

Table 1 shows the truth table of a binary full adder, where c_i, s, and c_o denote carry input, sum output, and carry out. Design a full adder using CMOS logic with PMOS and NMOS transistors.

Table 1: Truth table of a binary full adder.

x	y	c_i	s	c_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1